

AI Medical Video Generator — Project Overview

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Purpose

The tool turns a medical topic into a short, downloadable explainer video: an AI avatar presents it, with an AI voice, lip sync and animated visuals. No video reaches doctors until a clinical reviewer approves it.

Role	What they do in the tool
Content team	Enter a topic and source material, generate the video in the Studio, submit it for review
Clinical reviewer	Watch the video, edit scene narration, then approve, request changes or reject
Doctor	Stream or download approved videos from the Library

The current build is a proof of concept (Angular 16 front end, Node 18 API). Everything runs on one machine, with no paid services required.

Features

Every part of the video is generated: script, voice, presenter, lip movement, visuals and subtitles.

Feature	What it does
AI script	Writes a scene-by-scene storyboard from a topic and optional source content: narration, visual type and on-screen data per scene
AI voice	Speaks each scene's narration; returns the audio plus the timing of every word and mouth shape

Feature	What it does
AI avatar	An animated presenter with four looks: Doctor (white coat), Consultant, Surgeon (scrubs), Presenter (blazer). It blinks, breathes, sways and nods with the voice
Lip sync	The avatar's mouth follows the actual sounds being spoken, not a generic open-close loop
Animated visuals	Six scene types: avatar, whiteboard, step diagram, side-by-side comparison, disease progression, animated anatomy (heart, lungs, brain, kidney, liver, stomach, blood vessels, cells)
Subtitles	Captions highlighted word by word, in time with the voice
Downloadable video	A single WebM file with picture and voice, at 720p or 1080p
Clinical review	Review queue with approve, request changes and reject; reviewers can correct narration; every decision is logged
Doctor library	Lists only approved videos; unapproved files cannot be streamed or downloaded

User journey

A video moves through three screens: Studio, then Review, then Library.

Studio (content team)

1. Enter the category, topic and target duration. Optionally paste source content, such as guideline text or notes.
2. Choose a visual style (AI decides, Whiteboard, Step diagram, Anatomy illustration, Medical animation, Doctor presentation), a voice, an avatar, lip sync, subtitles and resolution.
3. Click *Write script*. The storyboard appears as editable scenes; change narration or visual type as needed.
4. Click *Generate video*. The tool voices every scene, then records the video in real time while you watch.
5. Download the file, or click *Submit for review*.

Review queue (clinical reviewer)

1. Open a submission and watch the preview.
2. Correct any scene's narration and add notes.
3. Choose *Approve & publish*, *Request changes* or *Reject*. Changes-requested videos go back to the author to resubmit.

Library (doctors)

1. Browse approved videos, each showing who approved it and when.
2. Stream in the browser or download the file.

Setup and running

Two commands install it and one starts it; API keys are optional. Requires Node 18.17 or later.

```
npm install
npm run server:install
cp server/.env.example server/.env    # optional: add API keys
npm run dev                          # API on port 3000, app on port 4200
```

Open `http://localhost:4200` in Chrome or Edge.

Capability	With an API key	Without a key
Script	Claude (<code>ANTHROPIC_API_KEY</code>)	Built-in offline planner (generic wording)
Voice	ElevenLabs (<code>ELEVENLABS_API_KEY</code>), most precise lip sync	macOS built-in neural voices; server must run on a Mac

`GET /api/studio/capabilities` shows which engines are active. As of this writing, no keys are configured, so the offline planner and macOS voices are in use.

Limitations and next steps

The pipeline works end to end; the gaps are in realism, file format and production hosting.

Limitation	Recommended next step
The avatar is an animated illustration, not a photoreal human	Plug in HeyGen, D-ID or Synthesia for the presenter; script, voice and review stay as they are
Offline scripts are generic	Add <code>ANTHROPIC_API_KEY</code> so Claude writes content from the source material
Output is WebM, which some doctor devices (older iPhones) may not play	Convert approved videos to MP4 on the server with FFmpeg
Recording runs in real time in the browser; the tab must stay open	Move rendering to a server job for long or batch videos
Videos and records are stored as local files	Move to a database and cloud storage, add login and roles

Review safety gap: if a reviewer edits narration and approves in one step, the video file keeps the old audio. Approval should be blocked until the video is regenerated.

- ☐ Add API keys for Claude and ElevenLabs
- ☐ Choose a photoreal avatar provider
- ☐ Add MP4 conversion after approval